MATHEMATICAL MODEL

Reverse-engineering plant cell mechanical properties

From forward mechanics to Bayesian inference, identifiability and uncertainty propagation

1 Problem statement

We observe a plant cell (or tissue) through (i) 3D images that give geometry $\mathcal{G}$, (ii) mechanical probes such as indentation, and (iii) deformation under osmotic change. We want the unobserved mechanical parameters

$$\boldsymbol{\theta} = (E,\ P,\ h,\ \nu,\ \phi_f, \dots),$$

the wall modulus E, turgor P, wall thickness h, Poisson ratio ν and microfibril orientation ϕ_f, together with a defensible statement of how well the data determine them. The model has three layers:

$$\underbrace{\boldsymbol{y} = \mathcal{G}_{\text{obs}}(\boldsymbol{\theta}, \boldsymbol{x}) + \boldsymbol{\eta}}_{\text{observation model}}, \qquad \underbrace{\delta\Pi(\boldsymbol{u}; \boldsymbol{\theta}) = 0}_{\text{forward mechanics}}, \qquad \underbrace{p(\boldsymbol{\theta} \mid \boldsymbol{y}) \propto p(\boldsymbol{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})}_{\text{inference}}.$$

> Section 4 gives a reduced closed-form model that can be inverted by hand and used for design and sanity checks. Sections 2–3 give the full model it is a special case of. Section 7 shows, with a runnable script, which parameters the data can and cannot determine.

2 Forward mechanics

2.1 Kinematics and variational principle

Let a wall segment have reference mid-surface $\mathcal{S}_w$ and thickness h_w, and let $\boldsymbol{F} = \nabla_s \boldsymbol{\chi}$ be the surface deformation gradient of the map $\boldsymbol{\chi} = \boldsymbol{X} + \boldsymbol{u}$. Cell c has volume $V_c(\boldsymbol{u})$ and turgor P_c. The equilibrium state is a stationary point of the potential energy

$$\Pi(\boldsymbol{u}; \boldsymbol{\theta}) = \sum_w \int_{\mathcal{S}_w} h_w\, \psi_w(\boldsymbol{F}; \boldsymbol{\theta}_w)\, dA \;-\; \sum_c P_c \left(V_c(\boldsymbol{u}) - V_c(\boldsymbol{0})\right), \tag{1}$$

with $\delta\Pi = 0$ for all admissible $\delta\boldsymbol{u}$. The pressure is a follower load, since $\delta V_c = \int_{\partial c} \boldsymbol{n} \cdot \delta\boldsymbol{u}\, da$. Shared walls between neighbours carry one stored energy, so tissue coupling enters through geometry and the shared degrees of freedom.

2.2 Wall constitutive laws

(a) Isotropic, small strain. Plane-stress Hooke law,

$$\boldsymbol{\sigma} = \frac{E}{1-\nu^2}\big[(1-\nu)\boldsymbol{\varepsilon} + \nu\,\mathrm{tr}(\boldsymbol{\varepsilon})\boldsymbol{I}\big]. \tag{2}$$

(b) Isotropic, finite strain. Incompressible neo-Hookean membrane, $\psi = \frac{\mu}{2}(I_1 - 3)$ with $\mu = E/3$.
(c) Fibre-reinforced, anisotropic. Two families of cellulose microfibrils with unit directions $\boldsymbol{a}_\pm = (\cos\phi_f, \pm\sin\phi_f)$ and $I_{4\pm} = \boldsymbol{a}_\pm \cdot \boldsymbol{C}\, \boldsymbol{a}_\pm$:

$$\psi = \frac{\mu}{2}(I_1 - 3) + \frac{\kappa_1}{2\kappa_2} \sum_{\pm} \Big(\exp\big[\kappa_2 \langle I_{4\pm} - 1\rangle^2\big] - 1 \Big), \qquad \langle x \rangle = \max(x, 0). \tag{3}$$

Here μ is the matrix stiffness, κ_1 the fibre stiffness, κ_2 the strain-stiffening rate, and fibres bear load only in tension. Then $\boldsymbol{\theta}_w = (\mu, \kappa_1, \kappa_2, \phi_f, h)$.

2.3 Irreversible growth (optional extension)

Lockhart-type growth adds an irreversible stretch rate $\dot{\varepsilon}^{\mathrm{ir}} = \varphi \langle \sigma_{\mathrm{wall}} - Y \rangle$, with extensibility φ and yield threshold Y, and total strain $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^{\mathrm{el}} + \boldsymbol{\varepsilon}^{\mathrm{ir}}$. It is needed only when the experiment is slow compared with wall remodelling.

3 Observation model

3.1 Measured quantities

Datum	Symbol	Model prediction
Turgid indentation stiffness	k_1	$\mathrm{d}F/\mathrm{d}\delta$ at the $P > 0$ equilibrium of (1)
Plasmolysed stiffness	k_2	same, with $P = 0$
Osmotic strain on turgor loss	ε	$(R - R_0)/R_0$ between the two equilibria
Wall thickness (segmentation)	h_{img}	h plus resolution-dependent error
Cell radius or geometry	R_{img}	R plus segmentation error

3.2 Error model

Positive quantities get multiplicative (log-normal) noise, $\ln y_j = \ln G_j(\boldsymbol{\theta}) + \eta_j$ with $\eta_j \sim \mathcal{N}(0, \sigma_j^2)$, so σ_j is a relative standard deviation. Wall thickness near the resolution limit has a systematic as well as a random error. Modelling the image as blurred by a point-spread function of width s gives

$$h_{\mathrm{img}} \approx \sqrt{h^2 + s^2}\, e^{\eta_h}, \tag{4}$$

so a bias correction, not just a variance, is needed when $h \lesssim s$. (This refinement is not used in the numerical example below.)

4 Reduced closed-form model

Replace the finite-element solve by thin-shell formulas for a nearly spherical cell of radius R, which is the regime of Section 7.

$$\text{Laplace law:} \quad \sigma = \frac{PR}{2h}, \qquad \varepsilon = \frac{(1-\nu)\,PR}{2\,E\,h}, \tag{5}$$

$$\text{Wall bending stiffness:} \quad k_{\mathrm{w}} = c\,Eh^2, \qquad c = \frac{4}{R\sqrt{3(1-\nu^2)}}, \tag{6}$$

$$\text{Turgor stiffness:} \quad k_{\mathrm{t}} = \alpha\,\pi\,PR, \tag{7}$$

$$\text{Observables:} \quad k_1 = k_{\mathrm{w}} + k_{\mathrm{t}}, \qquad k_2 = k_{\mathrm{w}}. \tag{8}$$

Equation (6) is the Reissner point-load result for a thin spherical shell. In (7), α is a probe-dependent geometry factor that in practice must be calibrated by FE or on a reference object; here $\alpha = 0.1$. These are surrogate relations, to be replaced by the FE model of Section 2 for real analysis.

4.1 Algebraic inversion

Thickness known from imaging.

$$P = \frac{k_1 - k_2}{\alpha\pi R}, \qquad E = \frac{k_2}{c\,h^2}. \tag{9}$$

Thickness eliminated using the osmotic strain. Let $s \equiv Eh$. From (5) and $PR = (k_1 - k_2)/(\alpha\pi)$,

$$s = \frac{(1-\nu)(k_1 - k_2)}{2\alpha\pi\,\varepsilon}, \qquad h = \frac{k_2}{c\,s}, \qquad E = \frac{c\,s^2}{k_2}. \tag{10}$$

With k_1 alone, (8) is one equation in two unknowns (E, P) and the problem is non-identifiable: all pairs on the line $k_{\mathrm{w}}(E) + k_{\mathrm{t}}(P) = k_1$ fit equally well.

4.2 First-order uncertainty propagation

Take logs of (9): $\ln E = \ln k_2 - \ln 4 + \ln R + \frac{1}{2}\ln\big(3(1-\nu^2)\big) - 2\ln h$. For independent inputs with relative errors $\sigma_k, \sigma_h, \sigma_R$ and absolute error σ_ν,

$$\sigma^2_{\ln E} = \sigma_k^2 + 4\,\sigma_h^2 + \sigma_R^2 + \left(\frac{\nu\,\sigma_\nu}{1-\nu^2}\right)^2. \tag{11}$$

The factor 4 is the key design fact: thickness error enters squared, so a 10% error in h costs 20% in E. For turgor, $\sigma^2_{\ln P} = \dfrac{(\sigma_k k_1)^2 + (\sigma_k k_2)^2}{(k_1 - k_2)^2} + \sigma_R^2 + \sigma_\alpha^2$. For the ratio route (10), $\ln E = \ln c + 2\ln s - \ln k_2$, so errors in ε, α and $(1-\nu)$ also enter doubled, which can cancel the benefit of removing h.

5 Statistical inference

5.1 Posterior

With data $\boldsymbol{y}$ and parameters $\boldsymbol{\vartheta} = (\ln E, \ln P, \ln h, \ln R, \nu, \ln\alpha)$,

$$p(\boldsymbol{\vartheta} \mid \boldsymbol{y}) \;\propto\; \underbrace{\prod_j \mathcal{N}\big(\ln y_j;\; \ln G_j(\boldsymbol{\vartheta}),\; \sigma_j^2\big)}_{\text{likelihood}} \;\underbrace{p(h)\,p(R)\,p(\nu)\,p(\alpha)}_{\text{informative priors from imaging / calibration}} \;\underbrace{p(E)\,p(P)}_{\text{log-uniform, wide}}. \tag{12}$$

Imaging measurements of h and R enter as priors (equivalently, as extra data) with their segmentation uncertainty, so imaging error propagates into (E, P) automatically.

5.2 Identifiability via the information matrix

Stack the weighted log-residuals $r_j = [\ln G_j(\boldsymbol{\vartheta}) - \ln y_j]/\sigma_j$ and let $\boldsymbol{J} = \partial \boldsymbol{r}/\partial\boldsymbol{\vartheta}$. The Fisher information (Gauss–Newton) matrix and Laplace posterior covariance are

$$\boldsymbol{F} = \boldsymbol{J}^\top \boldsymbol{J}, \qquad \mathrm{Var}(\boldsymbol{\vartheta} \mid \boldsymbol{y}) \approx \boldsymbol{F}^{-1}. \tag{13}$$

Structural identifiability means $\operatorname{rank}\boldsymbol{F} = \dim\boldsymbol{\vartheta}$. A zero eigenvalue of $\boldsymbol{F}$ with eigenvector $\boldsymbol{v}$ means the data are unchanged along $\boldsymbol{v}$. **Practical identifiability** concerns the size of the smallest eigenvalue relative to the prior and noise.

5.3 Cell-to-cell variability (hierarchical model)

For cells $i = 1, \ldots, N$,

$$\ln E_i = \mu_E + b_i, \quad b_i \sim \mathcal{N}(0, \tau_E^2), \qquad \ln P_i = \mu_P + b'_i, \quad b'_i \sim \mathcal{N}(0, \tau_P^2), \tag{14}$$

with weakly informative hyperpriors on (μ, τ). Biological spread τ must not be confused with measurement error: approximately $\hat\tau^2 \approx \mathrm{Var}_i(\ln \hat E_i) - \overline{\sigma_i^2}$, where σ_i^2 is the per-cell posterior variance from the previous subsection.

6 Computation

1. Segment images → geometry and its error model; mesh cells and walls.
2. Solve $\delta\Pi = 0$ by Newton iteration; obtain k_1, k_2, ε for candidate $\boldsymbol{\theta}$.
3. Optimise (MAP) with Gauss–Newton or L-BFGS, with gradients from the adjoint of the FE solve.
4. Compute $\boldsymbol{F}$, then profile likelihoods for parameters with weak curvature.
5. Sample the posterior with adaptive Metropolis or Hamiltonian Monte Carlo; check $\hat R$ and effective sample size.

6. Validate on synthetic data with known parameters before touching real data; hold out one measurement and check prediction.

The companion script `plant_cell_inverse_model.py` implements Sections 4–5 for the reduced model.

7 Numerical verification (synthetic)

Setup. Truth $E = 20$ MPa, $P = 0.30$ MPa, $h = 0.40\,\mu$m, $R = 15\,\mu$m, $\nu = 0.30$, $\alpha = 0.10$. This gives noise-free data $k_1 = 1.930$ N/m, $k_2 = 0.516$ N/m, $\varepsilon = 0.197$. Noise: 5% on each of k_1, k_2, ε, R; 10% on h_{img}; $\nu = 0.30 \pm 0.05$; 10% on α. Three experimental designs: **S** (k_1 only), **A** (k_1, k_2), **B** (k_1, k_2, ε).

	rank $\boldsymbol{F}$	Predicted rel. s.d. (Laplace)			400 repeated experiments	
Design	(of 6)	E	P	h	E	P
S: k_1 only	5	not identifiable			–	
A: k_1, k_2	6	21.3%	13.2%	10.0%	21.5%	13.6%
B: k_1, k_2, ε	6	16.9%	9.2%	8.6%	17.5%	8.6%

Equation (11) gives 21.3% for design A, matching the information-matrix value, and the empirical spread over 400 repeated synthetic experiments agrees with the predictions to within sampling error.

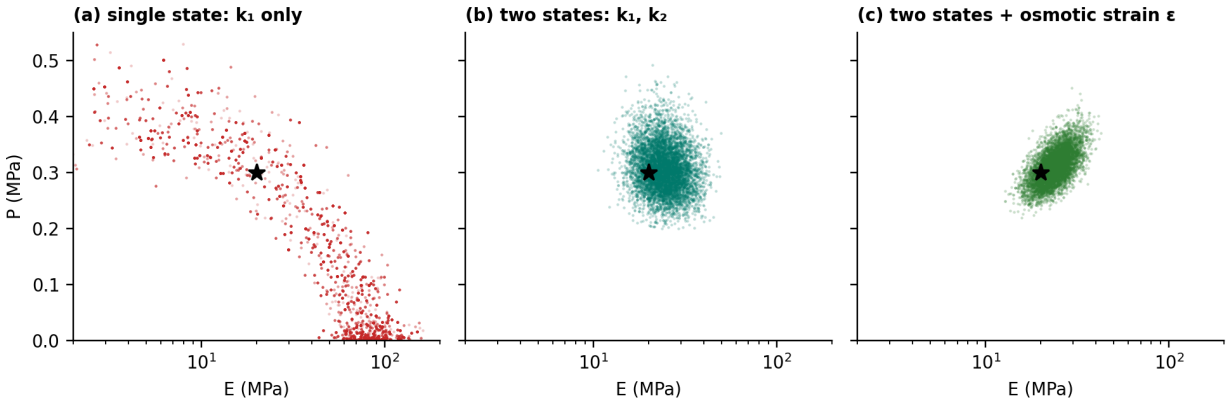


Figure 1. MCMC posterior of (E, P) for one synthetic dataset under each design. Star: truth. (a) The data constrain only a curved ridge; E ranges over two decades. (b) A second mechanical state closes the posterior. (c) Adding osmotic strain narrows it further.

MCMC, one dataset (95% interval)	E (MPa)	P (MPa)	h (μm)
S	0.1–117 (bounded by prior)	0.001–0.49	0.29–0.45
A	16.1–36.6	0.235–0.395	0.30–0.44
B	16.9–33.2	0.258–0.371	0.31–0.44

8 Interpretation and limits

- **A second mechanical state is what makes E and P separable.** A single turgid indentation cannot do it, whatever the noise level.
- **Thickness drives the error in E.** Because $E \propto h^{-2}$, image resolution and segmentation quality matter more than additional indentation repeats.
- **The osmotic strain helps, but is not free.** It adds information on the product Eh and shrinks all intervals, yet it also adds dependence on α and ν. Calibrating α with FE on the real probe geometry is a prerequisite.

- **Surrogate caveat.** Equations (6)–(7) treat the cell as a smooth sphere, the stiffness terms as additive, and strains as small. Real tissue shows strain stiffening, anisotropy, wall-thickness gradients and neighbour coupling, so model-form error can exceed noise. Test alternatives (Section 2.2) and compare with held-out data.
- The values in Section 7 are synthetic illustrations of the method. They are not measurements of any real cell.

Symbols

E	wall modulus (Pa)	k_1, k_2	turgid / plasmolysed indentation stiffness (N/m)
P	turgor pressure (Pa)	ε	osmotic (equibiaxial) strain
h	wall thickness (m)	α	probe geometry factor
R	cell radius (m)	$\boldsymbol{F}$	Fisher information matrix
ν	Poisson ratio	ϕ_f	microfibril angle
μ, κ_1, κ_2	matrix and fibre constants	τ	between-cell spread (log scale)